

PHYSICS MASTER SYLLABUS

CLASS 12 • NCERT / CBSE

Chapter 1: Electric Charges and Fields

- Electric Charge
- Properties of Electric Charge
- Coulomb's Law
- Superposition Principle
- Electric Field
- Electric Field Lines
- Electric Dipole
- Dipole in External Electric Field
- Electric Flux
- Gauss's Law
- Applications of Gauss's Law

Chapter 2: Electrostatic Potential and Capacitance

- Electrostatic Potential
- Potential Difference
- Equipotential Surfaces
- Potential Due to Point Charge
- Potential Due to Electric Dipole
- Potential Energy
- Conductors in Electrostatic Equilibrium
- Capacitor
- Capacitance
- Parallel Plate Capacitor
- Combination of Capacitors
- Energy Stored in Capacitor
- Dielectrics

Chapter 3: Current Electricity

- Electric Current
- Electric Charge Flow
- Drift Velocity
- Mobility
- Ohm's Law
- Resistance
- Resistivity
- Conductivity
- Temperature Dependence of Resistance
- Electrical Energy
- Electrical Power
- EMF
- Internal Resistance
- Cells in Series and Parallel
- Kirchhoff's Laws
- Wheatstone Bridge
- Meter Bridge
- Potentiometer

Chapter 4: Moving Charges and Magnetism

- Magnetic Force
- Lorentz Force
- Motion in Magnetic Field
- Cyclotron

- Biot-Savart Law
- Ampere's Circuital Law
- Magnetic Field Due to Current
- Solenoid
- Toroid

- Force Between Parallel Conductors
- Torque on Current Loop
- Moving Coil Galvanometer
- Ammeter
- Voltmeter

Chapter 5: Magnetism and Matter

- Bar Magnet
- Magnetic Field Lines
- Magnetic Dipole
- Earth's Magnetism
- Magnetic Elements
- Magnetisation

- Magnetic Intensity
- Magnetic Susceptibility
- Magnetic Permeability
- Diamagnetic Materials
- Paramagnetic Materials
- Ferromagnetic Materials

Chapter 6: Electromagnetic Induction

- Electromagnetic Induction
- Magnetic Flux
- Faraday's Laws
- Lenz's Law

- Motional EMF
- Eddy Currents
- Self Inductance
- Mutual Inductance

Chapter 7: Alternating Current

- Alternating Current
- Alternating Voltage
- RMS Value
- AC Circuit
- Phasor Diagram
- Resistance in AC Circuit
- Inductance

- Capacitance
- Reactance
- Impedance
- Series LCR Circuit
- Resonance
- Power in AC Circuit
- Transformer

Chapter 8: Electromagnetic Waves

- Displacement Current
- Electromagnetic Waves

- Properties of Electromagnetic Waves
- Electromagnetic Spectrum

- Applications of Electromagnetic Waves

Chapter 9: Ray Optics and Optical Instruments

- Reflection of Light
- Reflection at Spherical Surfaces
- Spherical Mirrors
- Mirror Formula
- Refraction of Light
- Refraction at Plane Surface
- Refraction at Spherical Surface
- Total Internal Reflection
- Optical Fibres
- Lenses
- Lens Formula
- Power of Lens
- Lens Combination
- Refraction Through Prism
- Microscope
- Astronomical Telescope

Chapter 10: Wave Optics

- Wave Nature of Light
- Huygens' Principle
- Wavefront
- Interference
- Young's Double Slit Experiment
- Diffraction
- Polarisation

Chapter 11: Dual Nature of Radiation and Matter

- Photoelectric Effect
- Hertz Experiment
- Einstein's Photoelectric Equation
- Photon
- Matter Waves
- de Broglie Hypothesis
- Davisson-Germer Experiment

Chapter 12: Atoms

- Atomic Models
- Thomson Model
- Rutherford Model
- Bohr Model
- Hydrogen Spectrum
- Energy Levels
- Atomic Excitation
- Atomic De-excitation

Chapter 13: Nuclei

- Atomic Nucleus
- Nuclear Composition
- Nuclear Size
- Nuclear Density
- Mass Defect
- Binding Energy
- Nuclear Stability
- Radioactivity
- Radioactive Decay Law
- Half-life
- Mean Life
- Nuclear Fission
- Nuclear Fusion

Chapter 14: Semiconductor Electronics: Materials, Devices and Simple Circuits

- Semiconductor Materials
- Intrinsic Semiconductors
- Extrinsic Semiconductors
- p-Type Semiconductor
- n-Type Semiconductor
- p-n Junction
- Semiconductor Diode
- Rectifier
- Zener Diode
- Light Emitting Diode
- Photodiode
- Solar Cell
- Junction Transistor
- Transistor Configurations
- Logic Gates
- Digital Circuits